Data Mining Project Proposal Submission

Group Members:
Taha Nawaz – 23L-2644, Ali Ahmad – 23L-2619 & Shahzeb Imran – 23L-2506

Topic:

Time-Series Data Analysis and Trend Discovery in Pakistan's Crop Prices

Our objective is to apply systematic time-series mining techniques to identify seasonal patterns, long-term trends, and abnormal fluctuations in agricultural price data.

Dataset Description:

The selected dataset is the **Crop Prices Dataset of Pakistan**, available on Kaggle. It contains historical crop price records collected from different agricultural markets across Pakistan. The dataset includes attributes such as crop type, market location, date, and price observations recorded over time.

This dataset naturally aligns with the domain of **Time-Series Data Analysis and Trend Discovery**, as price values are indexed temporally. The agricultural sector plays a critical role in Pakistan's economy, and crop prices are influenced by seasonality, supply-demand cycles, climatic factors, and market dynamics. Therefore, identifying long-term trends, seasonal effects, and volatility patterns is both academically meaningful and practically significant.

The core problem we aim to address is:

- *Can we systematically discover and model seasonal patterns, long-term trends, and abnormal fluctuations in Pakistan's crop prices using time-series data mining techniques?*

By answering this question, we aim to uncover interpretable temporal insights that may support better agricultural planning and market forecasting.

Dataset Link:
https://www.kaggle.com/datasets/humairarana/crop-prices-dataset-of-pakistan

Project Structure:

1. **Time-Series Preprocessing**
   a. Date parsing and chronological ordering
   b. Handling missing values
   c. Outlier detection and smoothing
2. **Temporal Feature Representation with the Following Possible Algorithms**
   a. Rolling Mean / Moving Average

b. Seasonal Decomposition (Additive / Multiplicative)
c. Differencing for Stationarity

3. **Calculating Temporal Patterns and Dependencies**
   a. Autocorrelation Function (ACF)
   b. Partial Autocorrelation Function (PACF)
4. **Application of Time-Series Modeling Algorithms**
   a. ARIMA
   b. Holt-Winters Exponential Smoothing

This structured pipeline ensures proper transformation of raw chronological data into interpretable trend and seasonal components consistent with time-series data mining principles.

Evaluation Metrics:

To evaluate forecasting accuracy and reliability of discovered trends, we will use:

- **Mean Absolute Error (MAE)**
- **Root Mean Squared Error (RMSE)**
- **Mean Absolute Percentage Error (MAPE)**

Lower values indicate better model performance. Temporal train-test splitting will be applied to preserve chronological order and ensure valid time-series evaluation.

Plan of Work:

- **First Deliverable** will finalize problem formulation, perform initial preprocessing, and conduct exploratory time-series analysis to understand seasonal behavior and volatility.
- **Second Deliverable** will implement decomposition techniques and baseline forecasting models (ARIMA and Holt-Winters), followed by preliminary evaluation and comparison.
- **Final Deliverable** will refine models through tuning, perform comparative evaluation, analyze residual behavior, extract meaningful trends and seasonal insights, and compile the complete analytical report and presentation.

Tech Stack:

Python, Pandas + any required library that conforms to the rules of algorithm implementation.

Expected Contribution:

This project demonstrates the application of **Time-Series Data Analysis and Trend Discovery** on real-world agricultural data. By systematically modeling and evaluating crop price trends, the study aims to provide interpretable temporal insights while maintaining academic rigor and experimental validation aligned with university-level data mining standards.